CONFIDENTIAL

Faculty of Computing

UNIVERSITI TEKNOLOGI MALAYSIA

RESEARCH UNIVERSITY

FINAL EXAMINATION
SEMESTER II, 2013/2014

SUBJECT CODE	**:**	**SCSR2043 / SCR2043**
SUBJECT	**:**	**OPERATING SYSTEM**
YEARS / COURSE	**:**	**SCSI/J/R/V/D/B, SCI/J/R/V/D/B**
TIME	**:**	**2 HOURS 30 MINUTES**
DATE	**:**	
PLACE	**:**	

INSTRUCTIONS :

Answer **ALL** questions in SECTION A and SECTION B in the answer booklet.

(Please fill in your particulars)

Name	
I/C No.	
Years / Course	
Section	
Lecturers Name	

This questions paper consists of (11) printed pages, including this page.

SECTION A

TRUE & FALSE QUESTIONS (10 MARKS)

(**INSTRUCTION**: Write your answer either **TRUE (T)** or **FALSE (F)** based on the following statements.)

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1. Deadlock avoidance requires that the operating system be given in advance additional information concerning which resources a process will request and use during its lifetime. [T]
2. A system is unsafe if the system can allocate resources to each process (up to its maximum) in some order and still avoid a deadlock. [F]
3. For a program to be run it has to be just moved from disk to memory. [T]
4. Cache memory sits between main memory and CPU registers. [T]
5. Logical address space is defined by Limit Registers and Cache memory. [F]
6. In an Operating System, physical address space of a process must be contiguous. [F]
7. With Contiguous disk allocation method, each file block contains a pointer to the next block on the disk drive. [F]
8. RAID provides reliability with data mirroring. [T]
9. Segmented memory allocation scheme does solve internal fragmentation problem. [T]
10. The basic principal of page replacement algorithm specify that when the number of frames allocation increases, this could resulted with the number of page faults decreases. [T]

SECTION B

STRUCTURED QUESTIONS (50 MARKS)

(**INSTRUCTION**: Please answer all questions in the space provided.)

QUESTION 2 [20 marks]

(a) In an operating system, dynamic memory partition is implemented.

i) Show how the following set of jobs {Job1: 200 KB, Job2: 280 KB, Job3: 417 KB, and Job4: 500 KB} are allocated inside the main memory. [3 marks]

ii) Referring to Table 2, assume at t_4 Job3 ends and at t_6 Job1 ends producing the following new arrival sequence:

Table 2: Time and job description

Time	Job #	Job Size in KB
t_5	5	200
t_7	6	500

a) Show/draw how these new jobs are allocated into memory space at t_4 till t_7. [4 marks]

b) Is there any job that cannot be allocated into the memory? If yes, state which job and give a reason. [2 marks]

(b) Given new Jobs in Table 3 and main memory using relocatable dynamic partition in Table 4.

Table 3: Job description

Jobs	Size
1	40K
2	40K
3	20K
4	25K
5	10

Table 4: Memory partition

80K	
25K	
25K	
40K	
5K	

i) Draw and label each job that will be allocated into the memory and shades all unused partition clearly. [4 marks]

ii) Calculate the total of fragmentation of the memory for the above case. [2 marks]

i)If another Job 6 with size 35K arrives, can this job be allocated into the memory? Suggest your solution. [5 marks]

QUESTION 3 [8 marks]

(a) Given a paging system with 2^{24} bytes of physical memory; 2^{10} pages of logical address space and page size of 512 bytes.

i) What is the total size of the logical address space. [1 mark]

ii) What is the frame size of the physical memory. [1 mark]

iii) How many frames of the physical memory. [1 mark]

iv) How many number of entry in the page table. [1 mark]

v) Calculate the offset bits for a page of virtual memory. [1 mark]

vi) Draw the frame offset of the physical memory address and page offset of logical memory. [3 marks]

QUESTION 4 [15 marks]

(a) Given the following reference string:

`1 3 2 1 4 3 2 4 1 5 2 4`

For each of the following page replacement algorithms, calculate how many **page faults** will occur if **3 frames** are available:

i) Least Recently Used (LRU) [4 marks]

ii) Optimal Page Replacement (OPT) [4 marks]

iii) Based on your answer in a i) and a ii), which algorithm is better? Justify your answer. [2 marks]

(b) Define what is thrashing**?** [1 mark]

(c) State 2 techniques to prevent thrashing. [2 marks]

(d) Based on Figure 2, explain the upper and lower bound of page fault. [2 marks]

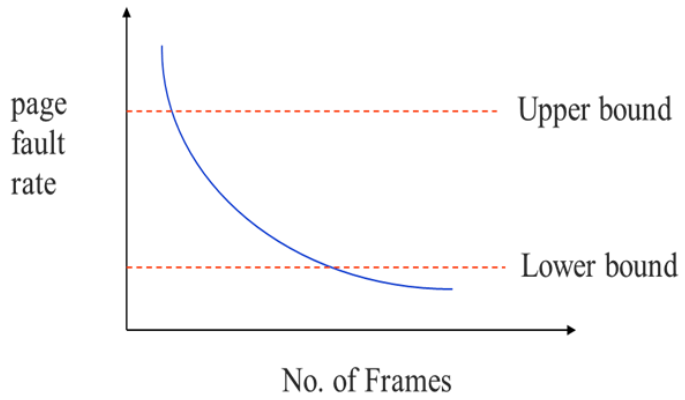


Figure 2: No. of frames vs page faulty rate

QUESTION 5 [24 Marks]

(a) Suppose a user of a UNIX system is currently in the home directory `/home/user1`. If the user issue the command `cd classes/scsr2043` and then the command `pwd`, what will be the user's current directory? Assume all directories exist. [1 mark]

(b) A user issues the Linux command `ls -l` and the following output is displayed on the screen:

```
-rw-r--r-- 1  muhalim  muhalim      0  Dec 30  13:12   fork2.c
drwxr-xr-x 2  muhalim  muhalim   4096  Dec 30  11:52   ResultC
```

i) Is `ResultC` a directory or a file? [1 mark]

ii) What is the access right of the owner of `fork2.c` [1 mark]

ii) Give a Linux command that will enable a group of users that share `fork2.c` to write to the file. [1 mark]

(c) Match the list of terms given below to each of the descriptions: [2 marks]

Access Control List	Access Control Matrix	Two-level directory
Contiguous Allocation	Indexed Allocation	Bit Map / Bit Vector
C-SCAN	SSTF	RAID	Data Stripping

i) A technique for spreading data over multiple disks.

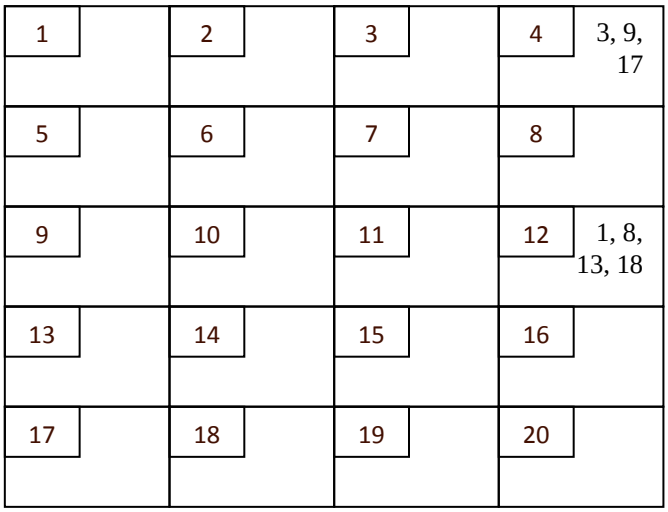
ii) A file protection mechanism that uses much more storage space as files and users increase.

iii) Provides easy direct access to records in a file (Every parts of a file are stored in same compact area).

iv) A technique to manage free spaces in disk.

(d) List 3 factors which determine how long it takes to read or write a disk block. Provide the formula to compute Access time. [2 marks]

(e) Figure 3 shows a diagram of disk storage containing two files, `File1` and `File2`:



File	Index block
File1	12
File2	4

Figure 3: Disk storage

i. Add arrows in the diagram if `File1` and `File2` start at block 12 and block 4 respectively. [3 marks]

ii. State the name of disk allocation technique that is being used. [1 mark]

(f) Figure 4 shows a diagram of disk storage containing two files, `File1` and `File2`:

Directory

File	Address	Size	Next
	1	512	
File1 (1)	2	512	8
File2 (1)	3	512	6
	. .	. .	
File2 (2)	6	512	11
	7	512	
File1 (2)	8	512	20
	. .	. .	
File2 (3)	11	512	-
	. .	. .	
File1 (3)	20	512	-

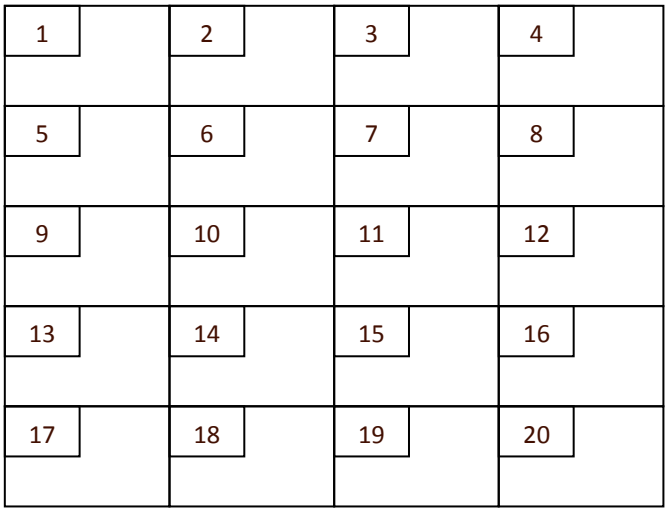


Figure 4: Disk Storage

i) Add arrows in the diagram according to the directory information for `File1` and `File2`. [3 marks]

ii) State the name of disk allocation technique that is being used. [1 mark]

(g) Suppose disk storage has block size of 4KB and disk size of 2 GB. Calculate total space needed for the bit vector. [2 marks]

(h) Assume the head of a disk is receiving track requests waiting in the queue in the following order:

98, 183, 65, 14, 122, 100, 20, 35

If the head starts at track 70 and going towards inner direction, show the movement path of the read-write head arm if the following seek strategies used:

i) C-SCAN [3 marks]

0 14 20 35 65 98 100 122 183 199

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ii) C-LOOK [3 marks]